# $u$-Substitution

## MATH 145 – Applied Calculus

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# The Fundamental Theorem of Calculus

Recall the Fundamental Theorem of Calculus from last lecture:

**The Fundamental Theorem of Calculus**

Suppose $f(t)$ is a continuous function on the interval $[a, b]$ and that $F(t)$ is an antiderivative of $f(t)$. Then

$$\int_a^b f(t)\,dt = F(b) - F(a)$$

The FTC says that you can evaluate $\displaystyle\int_a^b f(x)\,dx$ if you can find the antiderivative of $f(x)$. But how do we find $\displaystyle\int f(x)\,dx$?

# Rules for Integrals

There is no "Product Rule" for integrals, so while it's easy to find $\frac{d}{dx} f(x)g(x)$, it may not be as easy to find $\int f(x)g(x)$.

Remember that not all functions have clear antiderivatives, or antiderivatives which can be expressed in terms of elementary functions.

For example, the cumulative density function of the standard Normal distribution cannot be expressed easily and can generally only be found using numerical approximation methods.

$$\Phi(z) = \mathbb{P}[Z \leq z] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{\frac{t^2}{2}}\, dt$$

# Rules for Integrals

We can, however, use the Chain Rule to give us a trick to help solve stubborn integrals.

Recall the Chain Rule:

$$\frac{d}{dx} f(g(x)) = f'(g(x))\, g'(x)$$

Then we can have:

$$\int f'(g(x))\, g'(x)\, dx = \int \frac{d}{dx} f(g(x))\, dx = f(g(x))$$

**$u$-Substitution**

$$\int f'(g(x))\, g'(x)\, dx = \int f(u)\, du,$$

where

$$u = g(x).$$

# Example: $u$-Substitution

**Examples:**

- Find $\int 2xe^{x^2}\,dx$.

# Example: $u$-Substitution

**Examples:**

- Find $\displaystyle\int x^3(x^4+3)^2\,dx$.

- Find $\displaystyle\int x^3(x^2+3)^2\,dx$.

# Example: $u$-Substitution

**Examples:**

- Find $\displaystyle\int (x+1)\sqrt{3x+2}\,dx$.

- Find $\displaystyle\int \frac{1}{4x+3}\,dx$.

# Example: $u$-Substitution

**Examples:**

- Find $\int_0^{\frac{\pi}{2}} \sin^2(x)\cos(x)\,dx$.

- Find $\int_0^{\frac{\pi}{2}} \sin^2(x)\csc(x)\,dx$.

# $u$-Substitution for Definite Integrals

## $u$-Substitution for Definite Integrals

$$\int_a^b f'(g(x))\, g'(x)\, dx = \int_{g(a)}^{g(b)} f(u)\, du,$$

**Example:** Find $\displaystyle\int_0^1 e^{-x}(2e^{-x}+3)^9\, dx$.